

Syntax

- Terms refer to objects in a domain
- Predicates are used to make truth statements about objects

Functions

- $\mathcal{F}$ is the set of all functions.
- A function f of arity n map n terms to one term
- Constants are functions of arity 0

Terms

- functions $f \in \mathcal{F}$ refer to objects implicitly through other terms
- constants $c \in \mathcal{F}$ refer to individual objects
- variables $v \in \mathcal{V}$ are placeholders for objects

The set of terms $\mathcal{T}$ is defined inductively:

- $\forall v \in \mathcal{V} : v \in \mathcal{T}$
- $\forall c \in \mathcal{F} : c \in \mathcal{T}$
- $\forall f$ (of arity n) $\in \mathcal{F} : \forall t_1, \dots, t_n \in \mathcal{T} : f(t_1, \dots, t_n) \in \mathcal{T}$

Predicates

- $\mathcal{P}$ refers to the set of predicate symbols
- A predicate symbol $P \in \mathcal{P}$ of arity n maps n terms to an atom.
- A predicate symbol $P \in \mathcal{P}$ of arity 0 acts like a propositional variable.

Formulas

- The set of formulas is inductively defined as:
- If φ is a formula, then $\neg\varphi$ is a formula
- If φ, ψ are formulas, then $\varphi \odot \psi$ is a formula where $\odot \in \{\vee, \wedge, \rightarrow\}$
- If φ is a formula, then $\forall v \in \mathcal{V} : \mathcal{Q}x : \psi$ is a formula where $\mathcal{Q} \in \{\exists, \forall\}$

Ground Terms/Formulas

A term/formula is ground if it contains no variable.

Subformulas

- If φ is a atomic, then
 $subformulas(\varphi) = \{\varphi\}$
- If φ is of the form $\neg\psi, \forall x : \psi, \exists x : \psi$, then
 $subformulas(\varphi) = \{\varphi\} \cup subformulas(\psi)$
- If φ is of the from $\psi_1 \odot \psi_2$ where $\odot \in \{\vee, \wedge, \rightarrow\}$ then
 $subformulas(\varphi) = \{\varphi\} \cup subformulas(\psi_1) \cup subformulas(\psi_2)$

A formula $\psi \in subformulas(\varphi)$ is called a proper subformula of φ , if $\varphi \neq \psi$.

Binding Strength (to avoid parentheses)

1. $\forall, \exists, \neg$, 2. $\wedge, \vee$, 3. $\rightarrow$, 4. $\rightarrow$

Example :

$\neg A \wedge \exists x B \rightarrow C \vee D \leftrightarrow E$
is read as: $((\neg A) \wedge (\exists x B)) \rightarrow (C \vee D) \leftrightarrow E$

Quantifier scope and bound variables

Given a formula $\mathcal{Q}x : \psi$ where $\mathcal{Q} \in \{\exists, \forall\}$:
variable $x \in \mathcal{V}$ is bound in ψ
 ψ is the scope of x

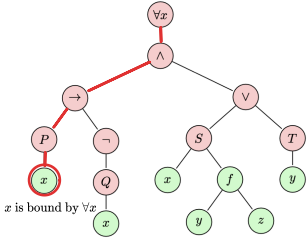
The quantifier only binds the first occurrence of a variable if outside a parentheses.

Example:

In the formula $\exists x P(x) \wedge R(x)$ only the first occurrence of x is bound, the second occurrence is free

Given a formula tree of a formula, a variable x is bound if it has an upward path to $\mathcal{Q}x$.

$\forall x((P(x) \rightarrow \neg Q(x)) \wedge (S(x, f(y, z)) \vee T(y)))$



Free variables

Given a term t , the set $free(t)$ of its free variables is defined as:

- If t is a variable $x \in \mathcal{V}$, then
 $free(t) = \{x\}$

- If t is a constant, then
 $free(t) = \emptyset$

- If t is of the form $f(t_1, \dots, t_n)$, then
 $free(t) = \bigcup_{i=1}^n free(t_i)$

Given a formula φ , the set $free(\varphi)$ of its free variables is defined as:

- If φ is an atom $P(t_1, \dots, t_n)$, then

$$free(\varphi) = \bigcup_{i=1}^n free(t_i)$$

- If $\varphi = \neg\psi$, then
 $free(\varphi) = free(\psi)$

- If $\varphi = \psi_1 \odot \psi_2$ where $\odot \in \{\vee, \wedge, \rightarrow\}$, then
 $free(\varphi) = free(\psi_1) \cup free(\psi_2)$

- If $\varphi = \mathcal{Q}x : \psi$ where $\mathcal{Q} \in \{\forall, \exists\}$, then
 $free(\varphi) = free(\psi) \setminus \{x\}$

Sentence

A formula is called a sentence if it contains no free variables.

Theory

A set $\Gamma = \{\varphi_1, \varphi_2, \dots\}$ of sentences is called a theory

Structures

A structure $\mathcal{M} = (\mathcal{D}, \mathcal{I})$ is made up of

- a domain $\mathcal{D}$
 - an interpretation function $\mathcal{I}$
- Constants, functions and predicates define the language (also called signature) $\mathcal{L}$ of the structure.

Domains

A domain $\mathcal{D}$ is a non-empty set of objects

Interpretation Function

- The interpretation function $\mathcal{I}$ maps:
- each constant c to an object $\mathcal{I}(c) = D$
- each function f of arity n to a total function $\mathcal{I}(f) = D^n \rightarrow D$
- each predicate symbol P of arity n to a relation $\mathcal{I}(P) \subseteq D^n$

One can also write $c^{\mathcal{M}}, f^{\mathcal{M}}, P^{\mathcal{M}}$ instead of $\mathcal{I}(\dots)$.

Satisfaction of ground formulas

For a structure $\mathcal{M} = (\mathcal{D}, \mathcal{I})$ and an atom $P(t_1, \dots, t_n)$ it holds:

- $\mathcal{M} \models P(t_1, \dots, t_n) \iff (t_1^{\mathcal{M}}, \dots, t_n^{\mathcal{M}}) \in P^{\mathcal{M}}$

Furthermore, for any ground formula φ it holds:

- $\mathcal{M} \models \neg\varphi \iff \neg(\mathcal{M} \models \varphi)$
- $\mathcal{M} \models \varphi \odot \psi \iff (\mathcal{M} \models \varphi) \odot (\mathcal{M} \models \psi) \quad \text{for } \odot \in \{\vee, \wedge, \rightarrow\}$

Given only a structure, one can only prove for ground formulas

Satisfaction of Formulas

Given an interpretation $(\mathcal{M}, \ell)$ the satisfaction of formulas is defined as:

- $(\mathcal{M}, \ell) \models P(t_1, \dots, t_n) \iff (t_1^{(\mathcal{M}, \ell)}, \dots, t_n^{(\mathcal{M}, \ell)}) \in P^{\mathcal{M}}$
- $(\mathcal{M}, \ell) \models \neg\varphi \iff \neg((\mathcal{M}, \ell) \models \varphi)$
- $(\mathcal{M}, \ell) \models \varphi \odot \psi \iff ((\mathcal{M}, \ell) \models \varphi) \odot ((\mathcal{M}, \ell) \models \psi)$
- $(\mathcal{M}, \ell) \models \mathcal{Q}x : \varphi \iff \mathcal{Q}x : ((\mathcal{M}, \ell) \models \varphi) \quad \text{for } \mathcal{Q} \in \{\exists, \forall\}$

One may also write:

- $(\mathcal{M}, \ell) \models \varphi \iff \neg((\mathcal{M}, \ell) \models \neg\varphi), (\mathcal{M}, \ell) \models \varphi \iff \mathcal{M} \models_{\ell} \varphi$

Satisfaction of Sentences

Theorem

Given a structure $\mathcal{M}$, a formula φ and variable assignments ℓ, ℓ' , it holds:
 $(\forall x \in free(\varphi) : \ell(x) = \ell'(x)) \implies (((\mathcal{M}, \ell) \models \varphi) \iff ((\mathcal{M}, \ell') \models \varphi))$

Given a structure $\mathcal{M}$ and a sentence φ , it holds:
 $\forall \ell, \ell' : ((\mathcal{M}, \ell) \models \varphi) \iff ((\mathcal{M}, \ell') \models \varphi)$

Satisfiability Definition

A formula φ is satisfiable $\iff \exists(\mathcal{M}, \ell) : (\mathcal{M}, \ell) \models \varphi$

Consistent Theory

A theory Γ is consistent $\iff \exists \mathcal{M} : \forall \varphi \in \Gamma : \mathcal{M} \models \varphi$

Theorem: $(\forall \varphi \in \Gamma : \mathcal{M} \models \varphi) \iff \left(\mathcal{M} \models \bigwedge_{\varphi \in \Gamma} \varphi \right)$

Tableau

Same inference rules as PL tableau but with additional rules:

$t : \forall x F \quad f : \exists x F \quad f : \forall x F \quad t : \exists x F$
 $t : F(t/x) \quad f : F(t/x) \quad f : F(c/x) \quad t : F(c/x)$

For any variable-free term $t \in \Sigma^{par} \cup \Sigma$ (usually picking a term) already in tableau branch.

Note that for every ground-term in the branch the rule $f : F(c/x)$ must be applied infinitely many constants added to the language of $\mathcal{M}$

$F(t/x)$ denotes the formula F where all occurrences of x are replaced by t .

Undecidability of FOL (example)

Due to the $t : \forall x F$ and $f : \exists x F$ rules the tableau may expand forever, as can be seen with the following example. A tableau is only guaranteed to terminate if the formula is a contradiction.

Given an interpretation $(\mathcal{M}, \ell)$ and the formula: $\forall x : P(x, g(x))$, the tableau would expand as follows:

$t : \forall x : P(x, g(x))$ // Applying the inference rule by replacing x for a term c
 $t : P(c, g(c))$ // since $g(c) = c'$ maps to a term, a new ground term has appeared in the branch for which this rule must be applied
 $t : P(c, g(c'))$ // once again, a new ground term $g(c') = c''$ has appeared in the branch. This will never stop

Example of a structure

- constants: $zero$
 - functions: $succ$ (arity 1)
 - predicates: $nonneg$ (arity 1), $equals$ (arity 2)
- $\mathcal{D} = \{a, aa, aaa, \dots\}$
 $zero^{\mathcal{M}} = a$
 $\forall d \in \mathcal{D} : succ^{\mathcal{M}}(d) = da$
 $nonneg^{\mathcal{M}} = \{aa, aaa, \dots\}$
 $equals^{\mathcal{M}} = \{(d, d) \mid d \in \mathcal{D}\}$

The structure satisfies the following ground atom:

$equals(aa, succ(a))$

since $succ(a) = aa$ and $(aa, aa) \in equals^{\mathcal{M}}$

FOL Interpretation

An interpretation is a pair $(\mathcal{M}, \ell)$.

An interpretation of a term $t^{(\mathcal{M}, \ell)}$ is defined as:

- $t^{(\mathcal{M}, \ell)} = \ell(t)$ if $t \in \mathcal{V}$
- $t^{(\mathcal{M}, \ell)} = f^{\mathcal{M}}(t_1^{(\mathcal{M}, \ell)}, \dots, t_n^{(\mathcal{M}, \ell)})$ if $t = f(t_1, \dots, t_n)$

Modelling natural language

- All As are B
 $\forall x : (A(x) \rightarrow B(x))$
- Some As are Bs
 $\exists x : (A(x) \wedge B(x))$
- No As are B
 $\neg \exists x : (A(x) \wedge B(x))$
- Some As are not Bs
 $\exists x : (A(x) \wedge \neg B(x))$

(Typically) adjectives and verbs are predicates; nouns are terms

$\neg \exists x : (A(x) \wedge \neg B(x))$

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Herbrand Interpretation

A Herbrand interpretation is an interpretation which maps functions to applications of functions on terms. A Herbrand universe is the set of all ground terms.

- Example:** (same idea as before)
 $\mathcal{D} = \{zero, succ(zero), succ(succ(zero)), \dots\}$
 $zero^{\mathcal{M}} = zero$
 $\dots$

Variable Assignment

Given a structure $\mathcal{M}$, for a variable assignment $\ell : \mathcal{V} \rightarrow \mathcal{D}$ holds:

- Given variable $x \in \mathcal{V}$ and object $d \in \mathcal{D}$:
 $\ell[d/x]$ is a variable assignment such that
 $\ell[d/x](y) = \begin{cases} d & \text{if } y = x \\ \ell(y) & \text{if } y \neq x \end{cases}$

Modelling natural language (Examples)

One vehicle passed all inspections:
 $\exists x : (\text{Vehicle}(x) \wedge \forall y : (\text{Inspection}(y) \rightarrow \text{Passed}(x, y)))$

There is one vehicle in every lane:
 $\exists y : (\text{Lane}(x) \rightarrow \exists y : (\text{Vehicle}(y) \wedge \text{InLane}(y, x)))$

All the children eat a sandwich
 $\forall x : (\text{Child}(x) \rightarrow \exists y : (\text{Sandwich}(y) \wedge \text{Eats}(x, y)))$

All the children are in the playground:
 $\exists y : (\text{Playground}(y) \wedge \forall x : (\text{Child}(x) \rightarrow \text{IsIn}(x, y)))$

The robot removes a block. Then it places a block on the table:

$\exists x_1 : (\text{Block}(x_1) \wedge \text{Removes}(\text{robot}, x_1)) \wedge \exists x_2 : (\text{Block}(x_2) \wedge \text{PlacesOnTable}(\text{robot}, x_2))$

Equivalences of quantified formulas

$\neg \forall x : \varphi \iff \exists x : \neg \varphi \quad \neg \exists x : \varphi \iff \forall x : \neg \varphi$
 $\forall x : \varphi \wedge \forall x : \psi \iff \forall x : (\varphi \wedge \psi) \quad \exists x : \varphi \vee \exists x : \psi \iff \exists x : (\varphi \vee \psi)$
 $\mathcal{Q}x : \mathcal{Q}y : \varphi \iff \mathcal{Q}y : \mathcal{Q}x : \varphi$

If x not free in ψ :

$\psi \odot \mathcal{Q}x : \varphi \iff \mathcal{Q}x : (\varphi \odot \psi) \quad \text{for } \mathcal{Q} \in \{\forall, \exists\}, \odot \in \{\vee, \wedge\}$

Negation Normal Form

A formula φ is in NNF if the only connectives are $\odot \in \{\vee, \wedge, \neg\}$ where $\neg$ is only used on variables

Prenex Normal Form

A formula $\varphi = \mathcal{Q}_1 x_1 : \dots \mathcal{Q}_n x_n : \psi$ is in PNF with:

- $\mathcal{Q}_i \in \{\forall, \exists\}$ for $i = 1, \dots, n$ (called quantifier prefix)
- ψ is a quantifier-free formula (called matrix)

Computing PNF

- Eliminate all occurrences of $\rightarrow$
- Move all negations inwards
- Rename all variables apart
- Move quantifiers to the front

Example of computing the PNF

Given the formula: $\forall x : (\exists y : R(x, y) \wedge \forall y : \neg S(x, y) \rightarrow \neg(\exists y : R(x, y) \wedge P))$

- Step 1:
 $\forall x : (\neg(\exists y : R(x, y) \wedge \forall y : S(x, y)) \vee \neg(\exists y : R(x, y) \wedge P))$
- Step 2:
 $\forall x : (\$